

DD: OSR-RMA-01 — Equipment Swap & RMA (parent)

Developer implementation guide: DD_OSR-RMA-01-Equipment_Swap_and_RMA-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract
DD_OSR-RMA-01-Equipment_Swap_and_RMA-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** Equipment Swap & RMA
Orchestration Parent DD for the OSR-RMA-01 constellation: framework + four flow satellites covering all equipment swap, pickup, and upgrade workflows for the operator. Provides the module overview, condensed per-flow narratives, satellite index, cross-DD coordination summary, and stub-debt plan. v1.0 scope: orchestration only. Per-flow BPMNs + per-flow tables + per-flow Kafka events live in the four satellite DDs. Shared schemas + workers + rules live in the framework.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/08_osr_equipment/DD_OSR-RMA-01-Equipment_Swap_and_RMA-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- PATCH /api/homepasses/{id}/network-path/ports
- POST /api/contractor-slot-commitments
- POST /api/equipment-instances/{id}/lifecycle-events
- POST /api/swap-requests/eqp
- POST /api/swap-requests/equ
- POST /api/swap-requests/gpon
- POST /api/swap-requests/hfc

- POST /api/swap-requests/{kind}
- POST /api/work-orders

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-charge-pre-authorize
- contractor-stock-route
- homepass-network-path-update
- instance-state-transition
- nms-remote-bind-stub
- nms-remote-unassign-stub
- noc-confirm-remote-success
- provisioning-adapter
- routing-decision-recovered-instance-routes-to-recovering-contractor
- serial-rebind-broadhub-stub
- serial-unbind-broadhub-stub
- tech-confirm-defect
- tech-confirm-recovery-or-eqr
- vendor-rma-coordinator-batch-ship

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.

4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** Equipment Swap & RMA Orchestration

Parent DD for the OSR-RMA-01 constellation: framework + four flow satellites covering all equipment swap, pickup, and upgrade workflows for the operator. Provides the module overview, condensed per-flow narratives, satellite index, cross-DD coordination summary, and stub-debt plan.

v1.0 scope: orchestration only. Per-flow BPMNs + per-flow tables + per-flow Kafka events live in the four satellite DDs. Shared schemas + workers + rules live in the framework.

1. What this module is

OSR-RMA-01 is the operator's home for every workflow where a serialized piece of equipment moves OUT of a customer's premises — whether it's being replaced (HFC swap), unbound remotely and reused (GPON remote swap), retrieved end-of-life (EQP for termination), or upgraded (EQU tech upgrade). The module sits on top of OSR-INSTANCE-01 (instance lifecycle) and OSR-01 (SKU+qty stock), orchestrating the per-instance state transitions and the contractor-routing of recovered units that those modules can't decide on their own.

The defining design constraint is that the four flows look 80% the same operationally — same Camunda primitives, same eligibility checks, same notification fanouts, same contractor-stock routing — but diverge sharply in their technology-specific provisioning (Broadhub slot/port preservation for HFC vs NMS remote unassign for GPON) and in their customer-facing intent (defect swap vs termination retrieval vs upgrade). The module split into framework + four satellites reflects that: shared infrastructure in one place, per-flow specifics each in their own document so a sprint team can pick up SWAP-HFC without dragging the full module weight into their context.

The module also solves a specific, documented operational pain: ~40+ stale equipment items in Wananchi KE today because unassigned recovered equipment auto-routes to the main warehouse instead of the contractor team's warehouse that physically retrieved it (workshop 2026-04-30, Customer Management deep dive). OSR-RMA-01 closes this gap by recording the recovering contractor on the swap-request AND on the per-instance lifecycle event, then routing the recovered unit back

to that contractor's van. The routing-decision-recovered-instance-routes-to-recovering-contractor Drools rule in the framework is the explicit codification.

What this module does

1. **Orchestrates four distinct swap flows** — SWAP-HFC, SWAP-GPON, EQP (Equipment Pickup), EQU (Equipment Upgrade). Each is a Camunda BPMN process with technology-specific service tasks, user tasks, and message catches. Per-flow BPMN definitions live in the satellite DDs; the framework provides the shared workers + Drools.
2. **Validates eligibility before committing field resources** — warranty status, customer billing state, subscription state, source/target SKU compatibility. Failed eligibility transitions the swap-request to FAILED with a specific failure code; no slot is reserved, no WO is created, no truck rolls.
3. **Reserves contractor capacity atomically** — calls EM-02 POST /api/contractor-slot-commitments with the resolved contractor for the customer's tech region. Slot is released on cancellation; capacity is freed.
4. **Creates and tracks Work Orders for field visits** — through WO-01 with new job types EQUIPMENT_SWAP, EQUIPMENT_PICKUP, EQUIPMENT_UPGRADE. WO finalization fires a message back into the swap-request BPMN.
5. **Drives per-instance lifecycle transitions** — every state change (source recovered, target placed, RMA shipped) calls OSR-INSTANCE-01 POST /api/equipment-instances/{id}/lifecycle-events. The contractor_id is carried on every recovery event.
6. **Routes recovered units to the recovering contractor's warehouse** — closes the documented 40+ stale-equipment gap. The contractor-stock-route worker reads recovery_contractor_id and writes both an OSR-01 stock_movement (SKU+qty) and an OSR-INSTANCE-01 lifecycle event (per-serial) in the same transaction.
7. **Calls provisioning adapters for OSS configuration** — HFC: Broadhub serial unbind/rebind with slot/port preservation (UC-PR4). GPON: NMS ONT unassign / bind. Adapters are operator-configurable stubs in v1.0; real OSS integration deferred.
8. **Applies charges or refunds via BIL-01** — chargeable swaps (out-of-warranty, customer-caused damage, upgrade fees) call BIL-01 on completion. EQP termination triggers deposit refund (or forfeiture in EQR case).
9. **Notifies customer and internal staff** — NOT-01 for customer-facing messages (appointment confirmation, completion, EQR equipment-return-required reminder); ICN-01 for internal candidate groups (swap-dispatchers, field-techs, noc-team).
10. **Captures vendor-RMA handoff metadata** — v1.0 STUB. When defective units are batched and shipped to vendor, the vendor_rma_stub row records the handoff. Per-vendor SLA tracking and return-batch matching are deferred to a future expansion.

What this module does NOT do

- **Doesn't orchestrate initial installation.** Installation is FUL-02's responsibility. OSR-RMA-01 takes over only when equipment is leaving the customer (or being upgraded in place).
- **Doesn't handle Trouble Tickets.** A ticket may trigger a swap (customer reports defect → ticket → ticket triage decides swap needed → ticket creates swap-request), but the ticket workflow lives in ASR-01 (pending). OSR-RMA-01 only handles the swap itself.
- **Doesn't make warranty decisions.** Warranty status comes from PLM-CFG-06 (SKU + warranty period) + the source instance's created_at / current_state history. The eligibility worker reads these and resolves to a chargeable-or-not flag; the actual charge code is from BIL-01's catalog. OSR-RMA-01 doesn't define warranty policy.
- **Doesn't run vendor portals.** The vendor_rma_stub is a back-office UI for ops to record what they shipped. Real-time vendor portal integration (vendor-issued reference numbers, vendor-side status updates, vendor-promised return dates) is deferred.
- **Doesn't allocate inventory.** When a swap needs a target instance, the swap-request's source caller (e.g., the satellite's POST endpoint) is responsible for resolving an available unit from OSR-INSTANCE-01 (filter current_state=IN_CONTRACTOR_STOCK for the recovery contractor, or IN_MAIN_WAREHOUSE for cross-contractor swaps). OSR-RMA-01 then sets that instance's state to RESERVED_FOR_WO. The allocation strategy (FIFO? LIFO? closest van?) is owned upstream.
- **Doesn't span operators.** A swap is operator-scoped. Cross-operator transfers (an instance moves from WIK to WUG warehouse) are out of v1.0 scope.

Regional flexibility

KE is the v1.0 seed; the four flows are configured per operator:

- **WIK** (HFC + GPON): all four satellites active. SWAP-HFC handles cable modems / STBs / MTAs / smart cards; SWAP-GPON handles ONTs; EQP handles both tech families on termination; EQU handles upgrades within and across tech families.
- **WUG, WTZ, YASSN** (GPON-only, pending workshop confirmation): SWAP-HFC satellite deployed but inactive (no HFC plant); SWAP-GPON + EQP + EQU active. Operator config table determines which satellites are exposed in the BO UI per operator.
- **Vendor-specific provisioning calls** are abstracted via the provisioning-adapter interface in the framework. WIK's adapter calls Broadhub + NMS at WIK URLs; YASSN's calls Free Sénégal's OSS endpoints. Same workers, different adapter configurations.

Glossary

Term	Meaning
Swap-request	One row in <code>equipment_swap_request</code> (framework). The workflow root.
Source instance	The OSR-INSTANCE-01 row representing the device being removed from field.
Target instance	The OSR-INSTANCE-01 row representing the device being placed in field (NULL for EQP).
Recovery contractor	The EM-02 contractor responsible for physical retrieval. Recorded on the swap-request and on the per-instance lifecycle event. The contractor whose warehouse the recovered unit routes back to.
Slot/port preservation	HFC-specific constraint (UC-PR4): the new modem must be authorized on the same CMTS slot/port as the old one.
EQR	Equipment Recovery failed — customer refused to return equipment during EQP. Closes swap with <code>final_state=COMPLETED_WITHOUT_RECOVERY</code> ; BIL-01 forfeits deposit.
Vendor RMA	Outbound shipment of a defective unit to its vendor for repair. v1.0 STUB only.
Provisioning adapter	The operator-specific abstraction over Broadhub / NMS / VoipSwitch / Verimatrix that the framework's workers call.

2. Constellation structure

2.1 Module breakdown

OSR-RMA-01 is split into a framework + four flow satellites:

File	Role	Lines
DD_OSR-RMA-01-Equipment_Swap_and_RMA-v1.0.md (this document)	Parent — module overview, glossary, condensed narratives, satellite index	~280
DD_OSR-RMA-01-FRAMEWORK-v1.0.md	Shared schemas (3), workers (14), Drools rules (5), user tasks (6), messages (8), FOUNDATION_AUTH groups (5)	~830
DD_OSR-RMA-01-FLOW-SWAP-HFC-v1.0.md	HFC field-pickup swap: BPMN, eligibility specifics (slot/port preservation), Broadhub double-config, pay-first for chargeable	~495
DD_OSR-RMA-01-FLOW-SWAP-GPON-v1.0.md	GPON swap: remote-first, field fallback, NMS unassign/bind, HomePass terminal-node update	~458
DD_OSR-RMA-01-FLOW-EQP-v1.0.md	Equipment Pickup at termination: paired with subscription closure; EQR sub-state handling, deposit refund vs forfeiture	~368
DD_OSR-RMA-01-FLOW-EQU-v1.0.md	Equipment Upgrade: chargeable always, deposit delta calculation, target-from-stock, tech-family-specific provisioning + HomePass for GPON	~401

2.2 Cross-document conventions

- Each satellite declares "**Satellite of: OSR-RMA-01**" in its header.
- **Schema additions in satellites are deltas only**. Each satellite may add ≤ 1 new table specific to its flow's metadata; the bulk of schema lives in the framework's `equipment_swap_request.flow_payload` JSONB.

- **BPMN files live with their satellite.** SWAP_HFC.bpmn in the SWAP-HFC satellite, SWAP_GPON.bpmn in the SWAP-GPON satellite, etc. The framework documents the workers + messages + user tasks but doesn't carry the BPMN diagrams.
- **REST endpoints are satellite-specific.** Each satellite declares its POST /api/swap-requests/{kind} create endpoint with the kind-appropriate flow_payload shape. Framework provides only the kind-agnostic GET / advance / cancel endpoints.
- **Kafka events are satellite-emitted, framework-cataloged.** Topics live on sophix.osr.rma.*; framework lists the events; satellites populate them.

3. Per-flow condensed narratives

3.1 SWAP-HFC — HFC field-pickup swap

When: Customer reports defect with HFC equipment (cable modem, MTA, STB, smart card), OR scheduled tech upgrade within HFC, OR proactive replacement by NOC.

Trigger: NOC agent / customer-care agent creates swap-request via satellite's POST /api/swap-requests/hfc. Source instance MUST be HFC-family (equipment_kind IN (CABLE_MODEM, MTA, STB, SMART_CARD)); target instance pre-resolved by caller.

Flow:

1. **Eligibility check** — warranty status, customer billing state, source instance currently IN_FIELD_ACTIVE or IN_FIELD_DEFECTIVE, target instance currently IN_CONTRACTOR_STOCK for the resolved contractor (or IN_MAIN_WAREHOUSE for cross-contractor). HFC-specific: source's network_slot_metadata.{cmts, slot, port} MUST be present; copied to swap-request flow_payload. Eligibility worker resolves chargeable flag + charge_code + customer_charge_amount (chargeable when warranty void / customer-caused damage).
2. **Pay-first gate (chargeable only)** — if chargeable=TRUE, worker bil01-charge-pre-authorize creates BIL-01 billable event; swap-request transitions to AWAITING_PAYMENT. Customer notified by NOT-01 with payment link. No slot reserved, no contractor dispatched until BIL-01 emits BillableEventPaid (Wananchi "wallet funding gate" — workshop 2026-04-28). Payment timeout (default 7 days) cancels the swap. Free swaps (warranty / system fault) skip this step.
3. **Slot reservation** — call EM-02 to reserve a contractor slot in the customer's tech region. recovery_contractor_id + slot_commitment_id populated.
4. **WO creation** — call WO-01 POST /api/work-orders with wo_kind=EQUIPMENT_SWAP_HFC, origin_ref=swap_request_id. WO is dispatched per WO-01-FLOW-INSTALLATION-like pattern (claim → dispatch → assign tech).
5. **Customer appointment confirmation** — NOT-01 sends customer SMS / email with appointment slot. Customer reply (accepted / rejected) routes via inbound webhook to the BPMN message catch.
6. **Field visit in progress** — tech arrives, calls tech-confirm-defect user task. Confirms defect; advances swap-request to FIELD_VISIT_IN_PROGRESS.
7. **Source recovery** — tech removes the old modem. Worker instance-state-transition advances source instance IN_FIELD_ACTIVE → RECOVERED_BY_CONTRACTOR with contractor_id set. Worker contractor-stock-route writes the stock_movement (van location) and per-instance lifecycle. THE ROUTING FIX HAPPENS HERE.
8. **Target placement** — tech installs the new modem on the SAME slot/port (UC-PR4). Worker advances target instance RESERVED_FOR_WO → IN_FIELD_ACTIVE bound to customer.
9. **Provisioning — Broadhub serial rebind** — worker serial-rebind-broadhub-stub calls the Broadhub adapter: unbind old CM-MAC (+ MTA-MAC if hybrid chassis), bind new CM-MAC at the same {cmts, slot, port}. For static-IP packages, double-config (Broadhub + Clearcable NOMS) — adapter handles both calls.
10. **Completion** — swap-request advances to COMPLETED. Payment has already been collected at step 2 (if chargeable). NOT-01 sends customer "swap complete" notification. WO finalized.

Failure paths: Broadhub rebind fails 3 attempts → FAILED, source rolled back to van (IN_CONTRACTOR_STOCK), target rolled back to van. Customer rejects appointment 3 times → FAILED with reason. Tech reports source not present at premises (customer already moved / damaged unrecoverably) → manual triage path.

Detail: see DD_OSR-RMA-01-FLOW-SWAP-HFC-v1.0.md.

3.2 SWAP-GPON — GPON swap (remote-first, field fallback)

When: Customer reports defect with GPON ONT, OR scheduled tech upgrade within GPON, OR proactive replacement.

Trigger: Same as HFC but on GPON-family source. Caller creates via satellite's POST `/api/swap-requests/gpon`.

Flow (remote-first happy path):

1. **Eligibility check** — same primitives as HFC, except no slot/port preservation requirement (GPON uses OLT auto-discovery, ONT serial is the only authentication signal). Source's `network_slot_metadata.{olt, frame, slot, port}` copied to `flow_payload` for traceability. Eligibility resolves chargeable + charge_code.
2. **Pay-first gate (chargeable only)** — same as SWAP-HFC step 2. `AWAITING_PAYMENT` → `AWAITING_SLOT` on `BillableEventPaid`.
3. **Remote unassign attempt** — worker `nms-remote-unassign-stub` calls the NMS adapter. If success, advance to `SOURCE_RECOVERED` (source instance `IN_FIELD_ACTIVE` → `IN_FIELD_DEFECTIVE` since the unit is physically still at premises but no longer in service). NOC user task `noc-confirm-remote-success` asks NOC to confirm customer plan for self-swap.
4. **Customer self-swap** — NOT-01 sends customer instructions + appointment for new ONT pickup at shop / courier delivery. Customer picks up new ONT and self-installs. Worker `nms-remote-bind-stub` binds the new ONT serial in NMS once customer reports it online.
5. **HomePass network-path update** — worker `homepass-network-path-update` calls `HomePass-CFG PATCH /api/homepasses/{id}/network-path/ports` updating the terminal ONT node's code from old serial to new ONT serial. Mandatory for GPON swaps (the `HomePass.network_path` terminal node is the system-of-record for which ONT serves the address).
6. **Old ONT recovery** — at next shop visit or courier pickup, old ONT collected. Worker `contractor-stock-route` runs with `recovery_contractor_id` set (the contractor who collected the old unit, even via shop). Source instance `IN_FIELD_DEFECTIVE` → `RECOVERED_BY_CONTRACTOR` → `IN_CONTRACTOR_STOCK`. Target `RESERVED_FOR_WO` → `IN_FIELD_ACTIVE` bound to customer.
7. **Completion** — same as HFC.

Flow (field fallback path): If customer can't / won't self-swap (NOC user task receives `customerSelfSwapConfirmed=false`):

- Advance to `SLOT_RESERVED` → `WO_CREATED` → `FIELD_VISIT_IN_PROGRESS`, same as HFC pattern thereafter.
- The remote-attempt timer + retry logic is in the BPMN; if 3 remote-attempt failures, automatic field fallback.

Detail: see `DD_OSR-RMA-01-FLOW-SWAP-GPON-v1.0.md`.

3.3 EQP — Equipment Pickup at termination

When: Subscription closure (SUB-WF-TERMINATE-01 finalization) triggers retrieval of all customer-bound equipment.

Trigger: SUB-WF-TERMINATE-01 calls POST `/api/swap-requests/eqp` for each instance currently bound to the terminating account. One EQP swap-request per instance (multiple if customer has ONT + STB + MTA).

Flow (happy path):

1. **Eligibility check** — source instance `IN_FIELD_ACTIVE` or `IN_FIELD_DEFECTIVE`, subscription `current_state=TERMINATING` or `TERMINATED`, deposit held on account (BIL-01 read).
2. **Slot reservation + WO creation** — same as field-visit pattern.
3. **Customer appointment** — NOT-01 customer notification: "we'll send a tech to retrieve your equipment on X". Customer reply or auto-confirm.
4. **Field visit** — tech arrives, customer hands over equipment. Worker advances source instance `IN_FIELD_ACTIVE` → `RECOVERED_BY_CONTRACTOR`. Routing-back via `contractor-stock-route`.
5. **Subscription closure completion** — fires message back to SUB-WF-TERMINATE-01 indicating equipment recovered. Termination workflow proceeds with deposit refund via BIL-01 + PLM-CFG-05 adjustment.

Flow (EQR sub-state — customer refuses return):

1. Same as above through field visit.
2. Tech `tech-confirm-recovery-or-eqr` user task: selects `CUSTOMER_REFUSED_RETURN`, adds notes.

3. Swap-request advances FIELD_VISIT_IN_PROGRESS → COMPLETED_WITHOUT_RECOVERY. Source instance NOT transitioned to RECOVERED_BY_CONTRACTOR — stays in IN_FIELD_DEFECTIVE.
4. SUB-WF-TERMINATE-01 receives "equipment unrecoverable" message. Triggers BIL-01 deposit forfeiture path. Source instance may be DECOMMISSIONED after a write-off review by ops.

Detail: see DD_OSR-RMA-01-FLOW-EQP-v1.0.md.

3.4 EQU — Equipment Upgrade

When: Customer-initiated tech upgrade (DOCSIS 3.0 → 3.1, ONT v1 → v2, STB → STB-4K), OR operator-initiated (e.g., end-of-life of an SKU forces mandatory replacement campaign).

Trigger: Sales / customer-care agent creates via POST /api/swap-requests/equ. Caller specifies upgrade_from_sku and upgrade_to_sku in flow_payload. Source = customer's current device; target = a pre-allocated upgrade unit from contractor stock.

Flow:

1. **Eligibility check** — source SKU compatible with upgrade-to SKU (PLM-CFG-06 cross-check via equipment_kind and operator-configured upgrade paths), customer billing state ACTIVE, no outstanding upgrade-fee unpaid, upgrade-fee resolved via BIL-01 catalog. Always chargeable: chargeable=TRUE, charge_code = upgrade-fee code, customer_charge_amount = upgrade fee + deposit delta (if any).
2. **Pay-first gate** — mandatory for EQU. Worker bil01-charge-pre-authorize creates BIL-01 billable event covering upgrade fee + deposit delta. Customer-care confirms with customer; customer pays via configured channels. AWAITING_PAYMENT → AWAITING_SLOT on BillableEventPaid. This codifies the Wananchi upgrade flow (workshop 2026-04-28): CC quotes price → customer pays → BO operates the change → WO scheduled.
3. **Slot reservation + WO creation** — same field-visit pattern with wo_kind=EQUIPMENT_UPGRADE.
4. **Field visit** — tech swaps old for new. Worker advances both instances; routing-back via contractor-stock-route. The OLD instance is NOT necessarily defective — typically returned to main warehouse for re-deployment after inspection.
5. **Provisioning** — same per-tech path (HFC: Broadhub rebind, slot/port preservation per UC-PR4 if HFC upgrade-to is HFC; GPON: NMS unassign + bind, **plus HomePass network_path terminal-node update for GPON upgrades**).
6. **Completion** — customer notified. Source instance returns to main warehouse for inspection; if good, re-enters IN_MAIN_WAREHOUSE pool.

Detail: see DD_OSR-RMA-01-FLOW-EQU-v1.0.md.

4. Cross-DD coordination summary

(Detailed in DD_OSR-RMA-01-FRAMEWORK-v1.0.md §7. Summary here for parent-level orientation.)

Upstream callers (write paths):

- NOC + customer support (manual swap creation)
- SUB-WF-TERMINATE-01 (EQP creation on termination)
- Sales / customer-care (EQU creation)
- Network monitoring (proactive SYSTEM_FAULT_PROACTIVE — future)
- BPMN workers (state advancement throughout flows)
- Vendor RMA coordinators (vendor_rma_stub population)

Downstream consumers (read paths + events):

- OSR-INSTANCE-01 (lifecycle event recording — via worker calls, not Kafka subscription)
- OSR-01 Stock Chain Engine (stock_movement recording — via worker calls)
- BIL-01 Charging Engine (charge application on completion; deposit handling)
- NOT-01 (customer notifications)
- ICN-01 (internal staff notifications)

- WO-01 (field visit orchestration)
- EM-04 Reporting (pending)

Sibling modules with read+write relationships:

- EM-02 (contractor lookup + slot reservation + release)
- PLM-CFG-06 (SKU + equipment_kind + warranty period)
- ILM-CFG-01 (account + subscription lookup; account.homepass_id binding for HomePass resolution)
- **BIL-01** (pay-first gate: billable event created at AWAITING_PAYMENT for chargeable swaps; deposit forfeiture on EQR; deposit refund on EQP successful return)
- **HomePass-CFG** (GPON swaps only: terminal-node ONT-serial PATCH at TARGET_PLACED — the HomePass.network_path stays in sync with the physical device serving the address)

Foundation dependencies:

- FOUNDATION_AUTH (group codes: swap-dispatchers, field-techs, noc-team, vendor-rma-coordinators, finance-warranty-approvers)
- FOUNDATION_CAMUNDA (engine for BPMN orchestration)
- FOUNDATION_DROOLS (rule evaluation for eligibility + routing decisions)
- FOUNDATION_KAFKA (event fanout to consumers)

5. Stub debt plan

Stub	Owner	Retired by	Plan
vendor_rma_stub (in framework)	OSR-RMA-01-FRAMEWORK	Future vendor-RMA DD (out of v1.0 scope)	Add vendor portal integration, vendor SLA tracking, return-batch matching, vendor-issued reference number validation. Retirement migrates the stub's rows + adds per-vendor SLA + status columns.
serial-unbind-broadhub-stub, serial-rebind-broadhub-stub workers	OSR-RMA-01-FRAMEWORK	Real Broadhub adapter (TBD per operator codebase cartography)	The "stub" label reflects v1.0 state: workers exist + are wired; real Broadhub call paths replace the stubbed implementations once the operator codebase exposes the right APIs.
nms-remote-unassign-stub, nms-remote-bind-stub workers	OSR-RMA-01-FRAMEWORK	Real NMS adapter (TBD per operator codebase cartography)	Same as Broadhub stubs.
vendor-rma-coordinator-batch - ship user task (declared but no BPMN consumer in v1.0)	OSR-RMA-01-FRAMEWORK	Future vendor-RMA workflow	The user task type is registered; a future BPMN exposes it. v1.0 ops use the framework's vendor_rma_stub REST endpoints directly without a BPMN.

This DD itself introduces no NEW stubs. All stubs above were declared in the framework; this parent just inventories them for visibility.

6. Implementation phasing (v1.0)

A suggested deploy order for ops:

Phase 1 — Framework + simplest flow (Sprint 1–2):

- Deploy framework schemas, workers, Drools, FOUNDATION_AUTH groups
- Deploy SWAP-GPON-REMOTE happy-path (no field fallback yet)
- Test end-to-end with a controlled defect scenario in a single tech region

Phase 2 — HFC swap (Sprint 3–4):

- Deploy SWAP-HFC satellite with Broadhub stub adapter

- Validate slot/port preservation rule via Drools
- Configure real Broadhub adapter per cartography findings (replaces stub)

Phase 3 — EQP termination flow (Sprint 5):

- Deploy EQP satellite
- Wire SUB-WF-TERMINATE-01 to create EQP swap-requests on termination
- Validate EQR sub-state handling with finance team

Phase 4 — EQU upgrade flow (Sprint 6):

- Deploy EQU satellite
- Wire BIL-01 upgrade-free charging path
- Validate deposit delta calculations with finance team

Phase 5 — Field fallback for GPON + vendor RMA stub UI (Sprint 7):

- Extend SWAP-GPON satellite with field-fallback path
- Deploy back-office UI for vendor_rma_stub CRUD operations
- Onboard NOC team on the swap-dispatchers FOUNDATION_AUTH group

Total: 7 sprints for v1.0 (matches BP-08 GAP analysis estimate of 4 + 3 sprints for G-08-1 + G-08-2).

7. Workshop sources

(Detailed in DD_OSR-RMA-01-FRAMEWORK-v1.0.md §8 and individual satellite §8 sections. Summary here.)

- **BP-08 Equipment Swap (RMA) GAP analysis** — primary workshop driver. HFC field-pickup vs GPON remote-unassign; UC-PR4 slot/port preservation; UC-PR9 serial unbind/rebind; contractor-stock traceability gap (~40 stale items). G-08-1 + G-08-2 addressed.
- **Workshop 2026-04-29 (Work Order & Field Operations deep dive)** — WO Job Types: EQU (Equipment Upgrade), EQP (Equipment Pickup), EQR (Equipment Recovery failed). Map to flow satellites.
- **Workshop 2026-04-30 (Customer Management deep dive)** — Equipment-routing gap explicit. Wananchi target: trace + route to recovering contractor.
- **Workshop 2026-04-30 (Provisioning deep dive)** — HFC modem authorization bound to slot/port (UC-PR4); static-IP HFC double config (Broadhub + Clearcable NOMS); static-IP GPON manual via DHCP console; voice MTA/IAD ports.
- **Workshop 2026-04-30 (Service Termination)** — Equipment dismantle as separate retrieval WO; agent contractor dispatched; refund via Invoice Adjustments.

TZ codebase coverage today: to be filled in after codebase review. The TZ codebase has equipment lookup + ticket-revamp + on-site action flow but no formal swap orchestration; assessment pending cartography.

Gap & V3 design rationale: OSR-RMA-01 family introduces explicit Camunda-orchestrated per-tech swap workflows, formal slot/port preservation, formal contractor-stock-routing, and an outbound-RMA stub for future expansion. Addresses both G-08-1 (workflow gap) and G-08-2 (contractor stock traceability gap).